17

Dusty plasmas

17.1 Introduction

Non-neutral plasmas had one less species than a conventional plasma (one species instead of two); dusty plasmas have one more species (three species instead of two). Not surprisingly, the addition of another species provides new freedoms, which give rise to new behaviors.

The third species in a dusty plasma, electrically charged dust grains, typically have a charge to mass ratio quite different from that of electrons or ions. Several methods for charging dust grains are possible. Electron bombardment is the usual means for charging laboratory dusty plasmas, but photoionization or radioactive decay could also be operative mechanisms and may be important for certain space and astrophysical situations. Photoionization would make dust grains positive because photoionization causes electrons to leave dust grains. Radioactive decay of dust grains would make the dust grains develop a polarity opposite to that of the particle emitted in the decay process, e.g., alpha particle emission by dust grains would make the dust grains negative.

We shall consider here only the typical laboratory dusty plasma situation where the plasma is weakly ionized and dust grain charging is due to electron bombardment. Negative charging occurs because the electrons, being much lighter than the ions and usually much hotter, have a much larger thermal velocity than the ions. As a result, when a dust grain is inserted into the plasma it is initially subject to a greater flux of impacting electrons than impacting ions, thereby causing the dust grain to become negatively charged. The negative charge deposited on the dust grain eventually becomes sufficiently large to repel incident electrons and thus attenuate the incoming electron flux. On the other hand, the negative charge on the dust grain accelerates incident ions thereby increasing the ion flux to the dust grain. The net charge on the dust grain reaches equilibrium when the electron and ion fluxes intercepting the dust grain become equal. This is a dynamic equilibrium because it involves a continuous flow of plasma to the dust grain.

Some kind of external source is thus required to replenish the plasma electrons and ions continuously, because otherwise all the plasma electrons and ions would eventually accumulate on the dust grain. The dust grain charging process is quite similar to the development of a floating potential on an insulated probe immersed in a plasma (see discussion of Eq. (2.133)).

17.2 Electron and ion current flow to a dust grain

Because a dust grain is much heavier than an electron or an ion, a dust grain acts as an effectively infinitely massive scattering center for electrons and ions colliding with it. In typical dusty plasma laboratory experiments the collisional mean free path l_{mfp} greatly exceeds the Debye length. This situation is sketched in Fig. 17.1 where a dust grain with radius r_d is surrounded by an imaginary sphere with diameter l_{mfp} and the Debye radius is much smaller than l_{mfp}. The nominal distance from the dust grain to the location of the incident electron's or ion's previous collision is λ_{mfp}, the collision mean free path. Thus, incident electrons and ions can be considered collisionless inside a sphere of radius λ_{mfp} centered about the dust grain. Since electrons and ions are collisionless inside the l_{mfp} sphere they have Keplerian orbits associated with the electrostatic central force produced by the charge on the dust grain. Because the dust grain is shielded by other particles at distances greater than a Debye length, the electrons and ions experience this central force only when inside a sphere having the nominal Debye radius.

In order to develop a model for dust grain charging, it is first necessary to determine the effective collision cross-sections for electrons and ions colliding with the dust grain. A useful benchmark reference for this calculation is the

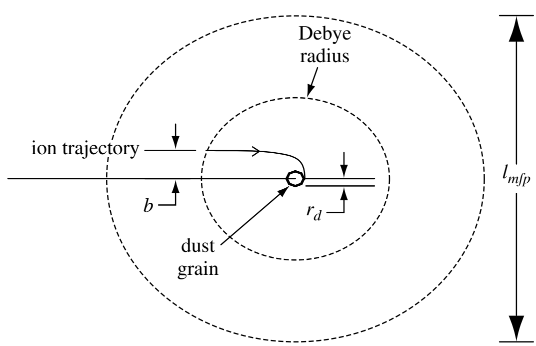

Fig. 17.1 Electrons and ions may be considered as being collisionless inside a sphere of diameter l_{mfp} surrounding a dust grain. The trajectory of an ion making a grazing collision is shown here; this ion has an impact parameter b.

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cross-section of charged particles colliding with a neutral dust grain. Because charged particles incident on a neutral dust grain have straight line trajectories, the cross-section of a neutral dust grain is just the geometrical area it projects onto a plane, i.e., $\sigma_{geometric} = \pi r_d^2$. The cross-section of a *charged* dust grain differs from $\sigma_{geometric}$ because incident electrons and ions are deflected from straight-line trajectories by the electrostatic central force produced by the charge on the dust grain. The respective cross-sections for ions and electrons will now be calculated and related to $\sigma_{geometric}$.

Figure 17.1 shows the impact parameter b and trajectory of an ion colliding with a negatively charged dust grain; the corresponding trajectory of an electron would curl outwards instead of inwards and so the electron would require a much smaller impact parameter b in order to hit the dust grain. We define v to be the initial velocity of an incident particle having mass m and charge q, and define v_{impact} as the velocity of this particle at the instant it makes a grazing impact with the dust grain. Using these definitions it is seen that conservation of angular momentum imposes the requirement

$$vb = v_{impact} r_d \tag{17.1}$$

and conservation of energy imposes the requirement

$$\frac{1}{2}mv^2 = \frac{1}{2}mv_{impact}^2 + q\phi_d, \tag{17.2}$$

where ϕ_d is the potential on the surface of the dust grain. Figure 17.1 shows that b is larger than r_d for an ion. Because the repulsive force between the negatively charged dust grain and an electron causes the trajectory of an electron to swerve in the opposite sense from an ion, b is smaller than r_d for an electron; that is, the electron has to be more "on-target" than a neutral particle to hit the dust grain, whereas an ion can be less "on-target" than a neutral particle to hit the dust grain.

Eliminating v_{impact} between Eqs. (17.1) and (17.2) gives

$$\frac{1}{2}mv^2 = \frac{1}{2}mv^2\frac{b^2}{r_d^2} + q\phi_d \tag{17.3}$$

so the effective scattering cross-section is (Allen, Boyd, and Reynolds 1957)

$$\sigma(v) = \pi b^2 = \left(1 - \frac{2q\phi_d}{mv^2}\right)\sigma_{geometric}. \tag{17.4}$$

For $q\phi_d > 0$ the interaction is repulsive and the cross-section is smaller than $\sigma_{geometric}$, whereas for $q\phi_d < 0$ the interaction is attractive and the cross-section is larger than $\sigma_{geometric}$. The former case applies to electrons and the latter case applies to ions.

The total current flowing to the dust grain for attractive interactions is

$$I_{attractive} = q \int_0^\infty \sigma(v) v f(v) 4\pi v^2 \mathrm{d}v. \tag{17.5}$$

However, because in the repulsive situation incident particles having $v < \sqrt{2q\phi_d/m}$ are reflected and do not hit the dust grain, the repulsive situation cross-section is zero for all particles having $v < \sqrt{2q\phi_d/m}$. Thus, the total current for repulsive interactions is

$$I_{repulsive} = q \int_{\sqrt{2q\phi_d/m}}^\infty \sigma(v) v f(v) 4\pi v^2 \mathrm{d}v. \tag{17.6}$$

Since the region outside the l_{mfp} sphere sketched in Fig. 17.1 extends to infinity, particles can be considered to have made many collisions before entering the l_{mfp} sphere, and so the velocity distribution of particles incident upon the l_{mfp} sphere will be Maxwellian, i.e.,

$$f(v) = \frac{n_0}{\pi^{3/2} v_T^3} \mathrm{e}^{-v^2/v_T^2}, \tag{17.7}$$

where $v_T = \sqrt{2\kappa T/m}$ is the thermal velocity. The incident particles will be collisionless as they travel inside the l_{mfp} sphere. Integration of Eqs. (17.5) and (17.6) gives

$$\begin{aligned} I_{attractive} &= q \int_0^\infty \pi r_d^2 \left(1 - \frac{2q\phi_d}{mv^2}\right) v f(v) 4\pi v^2 \mathrm{d}v \\ &= 2\pi^{1/2} r_d^2 n_0 q v_T \int_0^\infty \left(1 - \frac{2q\phi_d}{m v_T^2 x}\right) \mathrm{e}^{-x} x \mathrm{d}x \\ &= \frac{2}{\pi^{1/2}} \left(1 - \frac{q\phi_d}{\kappa T}\right) n_0 q v_T \sigma_{geometric} \end{aligned} \tag{17.8}$$

and

$$\begin{aligned} I_{repulsive} &= 2\pi^{1/2} r_d^2 n_0 q v_T \int_{q\phi_d/\kappa T}^\infty \left(1 - \frac{q\phi_d}{\kappa T x}\right) \mathrm{e}^{-x} x \mathrm{d}x \\ &= \frac{2}{\pi^{1/2}} \mathrm{e}^{-q\phi_d/\kappa T} n_0 q v_T \sigma_{geometric}. \end{aligned} \tag{17.9}$$

17.3 Dust charge

As discussed above, a neutral dust grain inserted into a plasma will become negatively charged because $v_{Te} \gg v_{Ti}$. The electron current I_e is thus a repulsive-type current and the ion current I_i is an attractive-type current. As the dust grain becomes more negatively charged, $|I_e|$ decreases and $|I_i|$ increases until $I_i + I_e = 0$, at which time dust grain charging ceases.

The presence of negatively charged dust grains affects the quasi-neutrality condition. Assuming singly charged ions and a charge of Z_d on the negatively charged dust grains, the quasi-neutrality condition generalizes to

$$n_{i0} = n_{e0} + Z_d n_{d0}. \tag{17.10}$$

Since the ion density is unaffected by the charging process, it is convenient to normalize all densities to the ion density and define

$$\alpha = \frac{Z_d n_{d0}}{n_{i0}} \tag{17.11}$$

so that

$$\frac{n_{e0}}{n_{i0}} = 1 - \alpha, \tag{17.12}$$

where n_{e0}, n_{i0}, and n_{d0} are the volume-averaged electron, ion, and dust grain densities. Furthermore, if r_d is small compared to the effective shielding length λ_d, then the dust grain charge Z_d is related to the dust grain surface potential and the dust grain radius r_d via the vacuum Coulomb relationship, i.e.,

$$\begin{aligned}\phi_d &= -\frac{Z_d e \exp\left(-r_d/\lambda_d\right)}{4\pi\varepsilon_0 r_d} \\ &\simeq -\frac{Z_d e}{4\pi\varepsilon_0 r_d} \text{ if } r_d \ll \lambda_d.\end{aligned} \tag{17.13}$$

Using Eq. (17.13) to give Z_d, Eq. (17.11) can then be written in terms of the dust grain surface potential as

$$\alpha = -\frac{4\pi\varepsilon_0 r_d n_{d0}}{n_{i0} e}\phi_d. \tag{17.14}$$

It is now convenient to introduce the dimensionless variable

$$\psi = -\frac{e\phi}{\kappa T_i} \tag{17.15}$$

so that α becomes

$$\alpha = 4\pi n_{d0} r_d \lambda_{di}^2 \psi_d, \tag{17.16}$$

where

$$\lambda_{di} = \sqrt{\frac{\varepsilon_0 \kappa T_i}{n_{i0} e^2}} \tag{17.17}$$

is the ion Debye length. Large ψ_d means that ions fall into a potential energy well much deeper than their thermal energy.

It is also useful to introduce the Wigner–Seitz radius a, a measure of the nominal spacing between adjacent dust grains. This spacing is defined by dividing the total volume of the system V by the total number of dust grains N to find a nominal volume surrounding each dust grain. It is then imagined that this nominal volume is spherical with radius a so that

$$N\frac{4\pi a^3}{3} = V, \tag{17.18}$$

in which case

$$n_{d0} = \frac{3}{4\pi a^3} \tag{17.19}$$

and

$$\alpha = \frac{3r_d\lambda_{di}^2}{a^3}\psi_d. \tag{17.20}$$

It is also convenient to normalize lengths to the ion Debye length so

$$\alpha = P\psi_d, \tag{17.21}$$

where

$$P = 3\frac{\bar{r}_d}{\bar{a}^3} \tag{17.22}$$

and the bar means normalized to an ion Debye length.

The dust charge can similarly be expressed in a non-dimensional fashion using Eq. (17.13) to give

$$\frac{Z_d}{4\pi n_{i0}\lambda_{di}^3} = \bar{r}_d\psi_d. \tag{17.23}$$

The floating condition $I_i + I_e = 0$ shows that the equilibrium dust potential must satisfy

$$0 = n_{i0}v_{T_i}\left(1 - \frac{e\phi_d}{\kappa T_i}\right) - n_{e0}v_{T_e}\exp\left(e\phi_d/\kappa T_e\right), \tag{17.24}$$

which can be rearranged as

$$(1+\psi_d)\sqrt{\frac{m_e T_i}{m_i T_e}}\exp\left(\psi_d T_i/T_e\right) = 1 - \alpha. \tag{17.25}$$

However, using Eq. (17.21) this becomes

$$(1+\psi_d)\sqrt{\frac{m_e T_i}{m_i T_e}}\exp\left(\psi_d T_i/T_e\right) = 1 - P\psi_d, \tag{17.26}$$

a transcendental equation relating ψ_d and P.

Equation (17.26) shows that, for a given temperature ratio T_i/T_e and a given mass ratio m_e/m_i, the normalized dust grain surface potential ψ_d is a function of

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P, a dimensionless parameter incorporating the geometrical information characterizing the dust grains in the dusty plasma. Because ψ_d appears non-algebraically in Eq. (17.26), it is not possible to solve for $\psi_d(P)$ without resorting to numerical methods. However, solving the inverse relation, namely $P(\psi_d)$, is straightforward because P occurs only once in Eq. (17.26). Solving for P gives (Havnes *et al.* 1987)

$$P = \frac{1}{\psi_d} - \left(1 + \frac{1}{\psi_d}\right)\sqrt{\frac{m_e T_i}{m_i T_e}} \exp\left(\psi_d T_i / T_e\right). \tag{17.27}$$

The upper plot in Fig. 17.2 shows $\log P$ plotted versus log ψ_d for two cases corresponding to typical laboratory configurations where dusty plasma experiments have been conducted; the lower plot shows the corresponding dependence of α on ψ_d. The $T_e = T_i$ configuration corresponds to a potassium Q-machine plasma while the $T_e = 100T_i$ configuration corresponds to an argon rf discharge. An ion of 40 atomic mass units has been assumed for both cases. Figure 17.2 shows that P varies inversely with ψ_d up to some critical value and then, for ψ_d above this critical value, P heads sharply to zero for modest increases in ψ_d. The lower plot shows that α is near unity to the left of the knee and then drops sharply to zero to the right of the knee. Comparison of the solid and dashed curves shows that the saturated value of ψ_d is an increasing function of T_e/T_i. The saturation value of ψ_d is 2.5 for the Q-machine parameters whereas the saturated

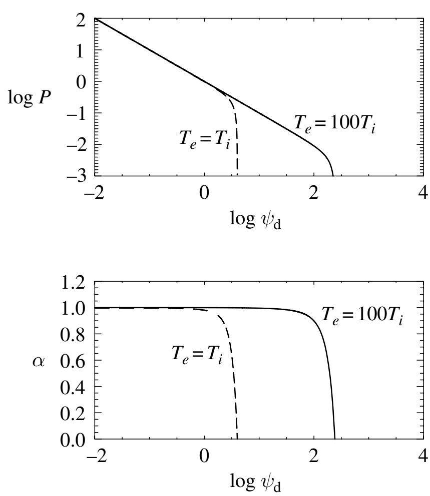

Fig. 17.2 Plots of $\log P$ and α vs. log ψ_d for $T_e = 100T_i$ and $T_e = T_i$. Logarithms are base 10 and ion mass is 40 amu. Because P is proportional to the dust grain density, the right-hand side of these curves, i.e., small P, corresponds to the limit of low dust grain density.

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value of ψ_d for the rf discharge is $\sim 2 \times 10^2$. Since P is proportional to $\bar{a}^{-3}$ and hence to the dust grain density, the right-hand side of these plots (i.e., where P is small) corresponds to the limit of small dust grain density. Since Eq. (17.23) shows that Z_d is proportional to ψ_d, it is seen that Z_d also saturates on moving to the right in the plots and largest Z_d occurs at high T_e/T_i and small dust grain density.

The downward sloping segment on the left of Fig. 17.2 corresponds to the $\alpha \simeq 1$ regime, which is the situation where the dust grain density is large and nearly all the electrons are attached to the dust grains so the plasma has almost no free electrons. On the other hand, the saturated limit of ψ_d on the right of Fig. 17.2 corresponds to the $\alpha \ll 1$ regime, which is where there is minimal depletion of free electrons, the dust grain density is low, and there is a very high charge on each of the relatively small number of dust grains.

There are thus three regimes:

1. *The regime well to the left of the knee in the curves.* Here $\alpha \simeq 1$ and nearly all electrons are attached to the dust grains. To the extent that α approaches unity, the dust grains replace the electrons as the negative charge carriers.
2. *The regime to the right of the knee in the curves.* Here $\alpha \ll 1$, most electrons are free, and the potential of an individual dust grain is very high and near its saturation value (right-hand side of plots). This regime corresponds to small P and a very small dust grain density.
3. *The regime in the vicinity of the knee in the curves.* If $T_e \gg T_i$ then it is possible to have α of order unity and ψ_d large, but not quite at its saturation value. In this regime the majority of electrons reside on the dust grains, the dust grains are highly charged, and the dust grain density is appreciable. Crystallization of the dust grains can occur in this regime, since crystallization requires a combination of high dust grain charge and small separation between dust grains.

17.4 Dusty plasma parameter space

A dusty plasma is characterized by the normalized dust radius $\bar{r}_d$ and the normalized dust interparticle spacing $\bar{a}$. These quantities determine P and, given P, the normalized dust surface potential ψ_d is found by solving Eq. (17.27). Knowing P and ψ_d then gives α using Eq. (17.21). The consequence of this chain of argument is that α can be considered to be a function of $\bar{r}_d$ and $\bar{a}$.

A dusty plasma parameter space can thus be constructed where $\bar{a}$ is the horizontal component and $\bar{r}_d$ is the vertical component so that a given dusty plasma would correspond to a point in this parameter space. The quantities ψ_d, α, and $Z_d/4\pi n_{i0}\lambda_{di}^3$ are all functions of $\bar{a}$ and $\bar{r}_d$ and so contours of these three quantities can be drawn in the $\bar{a}, \bar{r}_d$ parameter space for specified m_e/m_i and T_e/T_i.

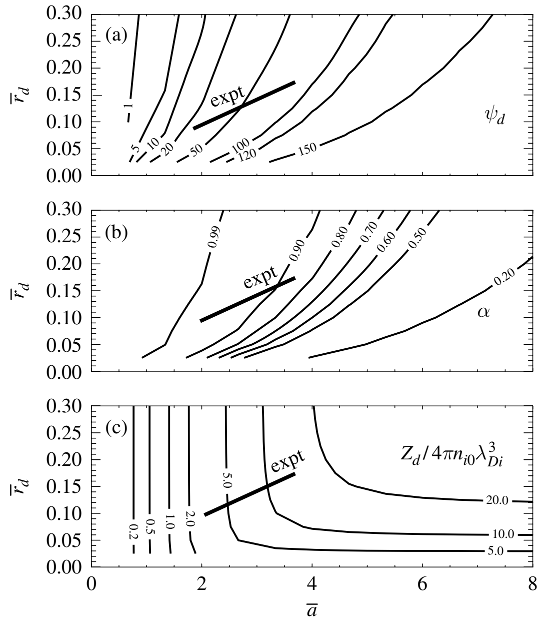

Fig. 17.3 Dusty plasma parameter space showing contours of constant ψ_d, α, and $Z_d/4\pi n_{i0}\lambda_{Di}^3$ for an argon plasma with $T_e/T_i = 100$. The experimental line corresponds to the density range of the Chu and I experiment (Chu and I 1994); these plots are from Bellan (2004b).

Examples of these contours are shown in Fig. 17.3 for an argon plasma with $T_e = 100T_i$. An actual experiment would have specific values of $\bar{a}$ and $\bar{r}_d$ and so would be represented as a point in this parameter space. Variation of the ion density in the experiment would change the value of λ_{di} while keeping the ratio $\bar{r}_d/\bar{a}$ fixed and so would correspond to moving along a sloped line in parameter space. The short sloped line in Fig. 17.3 represents the density in the dusty plasma experiment by Chu and I (1994) to be discussed later (the finite length of this line corresponds to the error bars for the density measurement).

17.5 Large *P* limit: dust acoustic waves

The large P limit (i.e., Regime 1 discussed above) has nearly all the electrons attached to the dust grains so that the plasma effectively consists of negatively charged dust grains and positive ions. A wave similar to the conventional ion acoustic wave can propagate in this regime, but the role played by positive and negative particles is reversed: here the ions are the light species and the dust grains are the heavy species. In order to appreciate the consequence of this role reversal, consider the conventional ion acoustic wave from the most simplistic point of view. The wave phase velocity ω/k is assumed to be much faster than the ion thermal velocity v_{Ti} (i.e., cold ion regime) but much slower than the electron thermal velocity v_{Te} (i.e., isothermal electron regime) so the respective approximations for the electron and ion equations of motion are

$$0 = -n_e e\mathbf{E} - \nabla\left(n_e \kappa T_e\right) \tag{17.28}$$

$$n_i m_i \frac{\mathrm{d}\mathbf{u}_i}{\mathrm{d}t} = n_i Ze\mathbf{E}. \tag{17.29}$$

Adding the electron and ion equations and assuming quasi-neutrality $n_e \simeq n_i Z$, gives

$$n_i m_i \frac{\mathrm{d}\mathbf{u}_i}{\mathrm{d}t} = -\nabla\left(n_e \kappa T_e\right) \tag{17.30}$$

showing that the effective force acting on the ions is the electron pressure gradient. This force is coupled to the ions via the electric field. In effect, the electron pressure gradient pushes against the electric field, which in turn pushes the ions. Linearizing Eq. (17.30) gives a result similar to a conventional sound wave, except that the system is isothermal with respect to the electron temperature (in a normal neutral gas sound wave, the gas temperature would appear in the right-hand side, the gas would be adiabatic, and so a γ would appear upon linearizing the gas pressure). Linearization of Eq. (17.30) and invoking the linearized quasi-neutrality relation $n_{e1} \simeq n_{i1} Z$, gives

$$n_{i0} m_i \frac{\partial \mathbf{u}_{i1}}{\partial t} = -Z\kappa T_e \nabla n_{i1}. \tag{17.31}$$

The linearized ion equation of continuity is

$$\frac{\partial n_{i1}}{\partial t} + n_{i0} \nabla \cdot \mathbf{u}_{i1} = 0. \tag{17.32}$$

The equations are combined by taking the divergence of Eq. (17.31) and then substituting Eq. (17.32) to obtain

$$\frac{\partial^2 n_{i1}}{\partial t^2} = \frac{Z\kappa T_e}{m_i} \nabla^2 n_{i1}, \tag{17.33}$$

which describes a wave with phase velocity $c_s^2 = Z\kappa T_e / m_i$.

This method can now be generalized to a plasma consisting of ions with charge $+e$, free electrons with charge $-e$, and dust grains with charge $-Z_d e$. The free electron density is assumed to be small so that $P \gg 1$, in which case the configuration is on the left of Fig. 17.2. The wave phase velocity is assumed to be much faster than the dust grain thermal velocity, but much slower than the ion/electron thermal velocities so the respective electron, ion, and dust grain equations of motion are

$$0 = -n_e e\mathbf{E} - \nabla\left(n_e \kappa T_e\right) \tag{17.34}$$

$$0 = +n_i e\mathbf{E} - \nabla\left(n_i \kappa T_i\right) \tag{17.35}$$

$$n_d m_d \frac{\mathrm{d}\mathbf{u}_d}{\mathrm{d}t} = -n_d Z_d e\mathbf{E}. \tag{17.36}$$

Adding the above three equations and invoking quasi-neutrality (i.e., $n_i = n_d Z_d + n_e$) results in

$$n_d m_d \frac{\mathrm{d}\mathbf{u}_d}{\mathrm{d}t} = -\nabla\left(n_i \kappa T_i + n_e \kappa T_e\right). \tag{17.37}$$

In analogy to the conventional ion acoustic wave, here the ion and electron pressures couple to the dust via the electric field. Linearization of Eq. (17.37) gives

$$n_{d0} m_d \frac{\partial \mathbf{u}_{d1}}{\partial t} = -\kappa T_i \nabla n_{i1} - \kappa T_e \nabla n_{e1}, \tag{17.38}$$

while linearization of Eqs. (17.34) and (17.35) gives

$$0 = -n_{e0} e\mathbf{E}_1 - \kappa T_e \nabla n_{e1} \tag{17.39}$$

$$0 = +n_{i0} e\mathbf{E}_1 - \kappa T_i \nabla n_{i1}. \tag{17.40}$$

Eliminating $\mathbf{E}_1$ between these last two equations shows that

$$\kappa T_e \nabla n_{e1} = -\frac{n_{e0}}{n_{i0}} \kappa T_i \nabla n_{i1}, \tag{17.41}$$

which can be integrated to give

$$n_{e1} = -\frac{n_{e0} T_i}{n_{i0} T_e} n_{i1}. \tag{17.42}$$

Inserting Eq. (17.42) into the linearized quasi-neutrality expression $n_{i1} = n_{d1} Z_d + n_{e1}$ gives

$$n_{i1} = \frac{1}{1 + \dfrac{n_{e0} T_i}{n_{i0} T_e}} Z_d n_{d1} \tag{17.43}$$

and hence

$$n_{e1} = -\frac{n_{e0} T_i / n_{i0} T_e}{1 + \dfrac{n_{e0} T_i}{n_{i0} T_e}} Z_d n_{d1}. \tag{17.44}$$

Substitution for n_{i1} and n_{e1} in Eq. (17.38) gives

$$n_{d0}m_d\frac{\partial \mathbf{u}_{d1}}{\partial t} = -\kappa T_i\frac{1-n_{e0}/n_{i0}}{1+\dfrac{n_{e0}T_i}{n_{i0}T_e}}Z_d\nabla n_{d1}$$

$$\simeq -\kappa T_i Z_d \nabla n_{d1} \tag{17.45}$$

since $n_{e0} \ll n_{i0}$ and typically $T_i/T_e \ll n_{i0}/n_{e0}$.

The linearized dust continuity equation is

$$\frac{\partial n_{d1}}{\partial t} + n_{d0}\nabla\cdot\mathbf{u}_{d1} = 0 \tag{17.46}$$

so taking the divergence of Eq. (17.45) and substituting Eq. (17.46) gives

$$\frac{\partial^2 n_{d1}}{\partial t^2} = \frac{Z_d\kappa T_i}{m_d}\nabla^2 n_{d1}, \tag{17.47}$$

which describes a wave with phase velocity

$$c_{da}^2 = \frac{Z_d\kappa T_i}{m_d}. \tag{17.48}$$

This wave is called the dust acoustic wave and its phase velocity is extremely low because of the large dust grain mass.

This analysis could be extended to include finite $k\lambda_D$ terms as obtained by using the full Poisson's equation instead of making the simplifying assumption of perfect neutrality. A Vlasov approach could also be invoked to demonstrate the effect of Landau damping. Rather than work through the details of these extensions, one can argue that the $\alpha \simeq 1$ dusty plasma is effectively a two-component plasma where the heavy particles are the negatively charged dust grains and the light particles are the positively charged ions. The previously derived results from both two-fluid theory and Vlasov theory could then be invoked by simply identifying the heavy and light particles in the manner stated above. Thus by making the identification shown in Table 17.1, taking into account finite $k^2\lambda_D^2$ terms will result in a dispersive dust acoustic wave

$$\omega^2 = \frac{k^2 c_{da}^2}{1+k^2\lambda_{Di}^2}. \tag{17.49}$$

Similarly, just as an electron flow velocity faster than the ion acoustic phase velocity would destabilize ion acoustic waves via inverse Landau damping, a Landau analysis would show that an ion flow velocity faster than the dust acoustic phase velocity would destabilize dust acoustic waves.

Destabilized dust acoustic waves have been observed in an experiment by Barkan, Merlino, and D'Angelo (1995) . The phase velocity of these waves was of

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Table 17.1

	ion acoustic wave	dust acoustic wave
inertia provided by	ions	dust grains
restoring force provided by	electron pressure	ion pressure

the order of 10 cm s^{-1}, which is an order of magnitude less than a typical human walking speed.

17.6 Dust ion acoustic waves

The presence of dust grains can also substantially modify the propagation properties of conventional ion acoustic waves. In a conventional electron–ion plasma, ion acoustic waves can propagate only if $T_e \gg T_i$. This is because strong ion Landau damping occurs when the wave phase velocity is of the order of the ion thermal velocity. This damping is impossible to avoid unless $T_e \gg T_i$, since the conventional ion acoustic phase velocity scales as $c_s^2 = (\gamma\kappa T_i + \kappa T_e)/m_i$.

As in the conventional ion acoustic wave, the derivation of dust ion acoustic waves involves the linearized electron and ion equations

$$0 = -n_{e0}e\mathbf{E}_1 - \kappa T_e \nabla n_{e1} \tag{17.50}$$

$$n_{i0}m_i\frac{\partial \mathbf{u}_i}{\partial t} = +n_{i0}e\mathbf{E}_1 - \gamma\kappa T_i \nabla n_{i1}; \tag{17.51}$$

the possibility of finite T_i has been retained in order to allow consideration of the $T_e \sim T_i$ regime. The wave frequency is assumed to be sufficiently high so that the dust grains are unable to respond to the wave. The dust grains can thus be considered as being infinitely massive and therefore stationary. In this limit the dust grains contribute to the equilibrium quasi-neutrality condition $n_{i0} = n_{e0} + Z_d n_{d0}$ but, since the dust grains are assumed stationary, their density cannot change and so the linearized quasi-neutrality condition is $n_{i1} = n_{e1}$. Thus, infinitely massive dust grains affect the equilibrium electron density, but not the perturbed electron density.

Eliminating $\mathbf{E}_1$ between Eqs. (17.50) and (17.51) gives

$$n_{i0}m_i\frac{\partial \mathbf{u}_i}{\partial t} = -\frac{n_{i0}}{n_{e0}}\kappa T_e \nabla n_{e1} - \gamma\kappa T_i \nabla n_{i1}. \tag{17.52}$$

Since $n_{i1} = n_{e1}$ and $n_{i0}/n_{e0} = n_{i0}/(n_{i0} - Z_d n_{d0})$, Eq. (17.52) becomes

$$n_i m_i\frac{\partial \mathbf{u}_i}{\partial t} = -\frac{1}{1 - Z_d n_{d0}/n_{i0}}\kappa T_e \nabla n_i - \gamma\kappa T_i \nabla n_i. \tag{17.53}$$

Taking the divergence and using Eqs. (17.11) and (17.32) gives

$$m_i \frac{\partial^2 n_{i1}}{\partial t^2} = \left(\frac{1}{1-\alpha} \kappa T_e + \gamma \kappa T_i \right) \nabla^2 n_{i1}, \tag{17.54}$$

which gives the dust ion acoustic wave (Shukla and Silin 1992), a wave with phase velocity

$$c_{DIA}^2 = \frac{1}{1-\alpha} \frac{\kappa T_e}{m_i} + \gamma \frac{\kappa T_i}{m_i}. \tag{17.55}$$

As α approaches unity, the dust ion acoustic wave phase velocity becomes much greater than the ion thermal velocity even in a plasma having $T_i = T_e$. The dust ion acoustic wave thus propagates without being attenuated by ion Landau damping in a $T_i = T_e$ plasma having $\alpha \simeq 1$, which corresponds to the left side of Fig. 17.2. Thus, the presence of a large dust grain density enables the propagation of ion acoustic waves that normally would be damped in a $T_e = T_i$ plasma; this has been observed in experiments by Barkan, D'Angelo, and Merlino (1996).

17.7 The strongly coupled regime: crystallization of a dusty plasma

The mutual repulsive force between two negatively charged dust grains scales as Z_d^2 and so will be very large for highly charged dust grains. In the extreme limit, the electrostatic potential energy between dust grains might exceed their kinetic energy so that the grains would tend to form an ordered, crystallized state. The possibility that dust grains might crystallize was first suggested by Ikezi (1986) and has since been demonstrated in a number of experiments (Chu and I 1994, Melzer, Trottenberg, and Piel 1994, Thomas *et al.* 1994, Hayashi and Tachibana 1994, Nefedov *et al.* 2003, Morfill *et al.* 2002). The threshold criterion for crystallization will be discussed in this section following a model by Bellan (2004b). The threshold is determined by considering certain issues relating to the validity of the conventional Debye shielding model and the Boltzmann relation.

Large Z_d corresponds to large ψ_d, which in turn corresponds to operating towards the right of Fig. 17.2. Since the location of the saturation value of ψ_d increases with T_e/T_i, very large ψ_d can occur if $T_e \gg T_i$. In addition to the scaling with Z_d^2 the repulsive force also scales inversely with the square of the distance separating the two dust grains, i.e., the repulsive force also scales as $n_{d0}^{2/3}$. Since P is proportional to n_{d0}, the maximum repulsive force would be obtained around the knee in the $T_e \gg T_i$ curves in Fig. 17.2 since at this location it is possible to have large ψ_d without n_{d0} becoming infinitesimal.

The repulsive electrostatic force between two dust grains is attenuated by Debye shielding. This shielding can be calculated by considering a single dust grain to

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be a test particle immersed in a plasma consisting of electrons, ions, and other dust grains (these other dust grains will be referred to as "field" dust grains). The test particle will be completely shielded beyond some critical radius. A field dust grain located beyond this critical radius will experience no interaction with the test particle dust grain whereas if the field dust grain is located within the shielding radius, it will experience an enormous repulsive force. The test particle dust grain thus acts like a finite-radius hard sphere in its interactions with field particle dust grains.

A quantitative model for these interactions between dust grains can be developed by considering Poisson's equation for a dusty plasma,

$$\nabla^2\phi = -\frac{1}{\varepsilon_0}\left(n_i e - n_e e - Z_d n_d e\right). \tag{17.56}$$

The usual test-particle argument (see p. 10) involves linearization of the Boltzmann relation for each species to obtain a linearized density for each species. These linearized densities are then substituted into Poisson's equation resulting in the Yukawa-type solution,

$$\phi = \frac{q_t}{4\pi\varepsilon_0 r}\exp\left(-r/\lambda_D\right), \tag{17.57}$$

where $\lambda_D^{-2} = \sum \lambda_{D\sigma}^{-2}$. However, this linearization is based on the assumption $|q_t\phi/\kappa T_\sigma| \ll 1$, which is clearly not true in the vicinity of a highly charged dust grain. This inconsistency cannot be resolved in fluid theory and it is necessary to revert to the more fundamental Vlasov description.

According to the Vlasov description, the phase-space density of particles is characterized using a velocity distribution function $f_\sigma(\mathbf{r}, \mathbf{v}, t)$ for each species and the time evolution of f_σ is prescribed by the Vlasov equation. In order to determine the time-averaged potential in the vicinity of a test particle, a steady-state solution to the Vlasov equation must be found. To do this, we begin by letting the test particle location define the origin of a spherical coordinate system and then assume that all time-averaged quantities are spherically symmetric about this origin so $\phi = \phi(r)$ and $f_\sigma = f_\sigma(r, v)$. Since an incident particle can be considered as colliding with the test particle, the nominal distance from the test particle to the location of the incident particle's previous collision is λ_{mfp}, the collision mean free path. Thus, so far as collisions with particles other than the test particle are concerned, an incident particle can be considered as being collisionless inside a sphere of radius λ_{mfp} centered about the test particle, i.e., centered about the origin. Because a typical particle incident upon the test particle will have traveled many mean free paths, the velocity distribution of incident particles will be Maxwellian when these particles enter a sphere with radius of order l_{mfp} centered on the test particle. Since the incident particles are collisionless within the λ_{mfp}

sphere, their velocity distribution function must be a solution to the collisionless Vlasov equation inside this sphere. The boundary condition this inside solution must satisfy is that its large r limit should correspond to the collisional (i.e., Maxwellian) distribution outside the λ_{mfp} sphere.

Solutions to the collisionless Vlasov equation are functions of constants of the motion as shown in Section. 2.3 and the appropriate constant of the motion here is the particle energy $W = m_\sigma v^2/2 + q_\sigma \phi(r)$. Hence, the distribution function in the collisionless region near the test particle must be

$$f_\sigma (r, \mathbf{v}) = n_{\sigma 0} \left(\frac{m_\sigma}{2\pi\kappa T_\sigma} \right)^{3/2} \exp\left(-\frac{m_\sigma v^2/2 + q_\sigma \phi(r)}{\kappa T_\sigma} \right), \tag{17.58}$$

since this maps to a Maxwellian distribution at large distances r, where $\phi(r) \to 0$.

A negatively charged particle such as a dust grain or an electron experiences a repulsive force upon approaching the dust grain test particle and so slows down. Some approaching negatively charged particles reflect and so the minimum velocity of electrons or dust grains approaching the dust grain test particle is zero. The density of these particles will thus be

$$\begin{aligned} n_\sigma &= \int_0^\infty f_\sigma (r, \mathbf{v})\, \mathrm{d}^3 v \\ &= n_{\sigma 0} \exp\left(-\frac{q_\sigma \phi(r)}{\kappa T_\sigma} \right), \end{aligned} \tag{17.59}$$

which is the same as the fluid theory Boltzmann relation. It is useful at this point to change over to the non-dimensional scalar ψ defined in Eq. (17.15). Since the dust grains are negatively charged, ψ is large and positive in the vicinity of a dust grain. The respective normalized electron and field dust grain densities are thus

$$\frac{n_e}{n_{e0}} = \exp\left(-\psi T_i/T_e\right) \tag{17.60}$$

$$\frac{n_d}{n_{d0}} = \exp\left(-\bar{Z}\psi\right), \tag{17.61}$$

where

$$\bar{Z} = Z_d T_i/T_d. \tag{17.62}$$

Both the electron and field dust grain densities decrease in the vicinity of the dust grain test particle. However, because $\bar{Z}$ is typically very large and T_i/T_e is assumed to be very small, the field dust density scale length is much shorter than the electron density scale length.

Ion dynamics are qualitatively different because all ions approaching the dust grain are accelerated, leading to the situation that no zero velocity ions exist near the dust grain. In particular, an ion starting with infinitesimal inward velocity at

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infinity where $\psi = 0$ has nearly zero energy, i.e., $W \simeq 0$. The energy conservation equation for this slowest ion is

$$m_i v^2/2 + e\phi(r) = 0 \tag{17.63}$$

and so the velocity for this slowest ion has the spatial dependence

$$v_{\min} = \sqrt{-\frac{2e\phi(r)}{m_i}}. \tag{17.64}$$

The ion density is thus

$$\begin{aligned} n_i(r) &= \int_{v_{\min}}^{\infty} f_i(r, \mathbf{v}) \,\mathrm{d}^3 v \\ &= n_{i0} \left(\frac{m_i}{2\pi\kappa T_i}\right)^{3/2} \exp\left(-\frac{q_i \phi(r)}{\kappa T_i}\right) \int_{v_{\min}}^{\infty} \exp\left(-\frac{mv^2/2}{\kappa T_i}\right) \mathrm{d}^3 v \\ &= n_{i0} \frac{4}{\sqrt{\pi}} \exp(\psi) \int_{\sqrt{\psi}}^{\infty} \exp\left(-\xi^2\right) \xi^2 \mathrm{d}\xi. \end{aligned} \tag{17.65}$$

This can be expressed in terms of the error function

$$\operatorname{erf} z = \frac{2}{\sqrt{\pi}} \int_0^z \exp(-\xi^2) \mathrm{d}\xi \tag{17.66}$$

and, in particular, using the identity

$$\frac{4}{\sqrt{\pi}} \int_z^{\infty} \exp\left(-\xi^2\right) \xi^2 \mathrm{d}\xi = \frac{2}{\sqrt{\pi}} \int_z^{\infty} \mathrm{d}\xi \exp(-\xi^2) - \frac{2}{\sqrt{\pi}} \int_z^{\infty} \mathrm{d}\xi \frac{\mathrm{d}}{\mathrm{d}\xi} \left(\xi \exp(-\xi^2)\right) \tag{17.67}$$

it is seen that (Laframboise and Parker 1973)

$$\frac{n_i}{n_{i0}} = \exp(\psi)\left(1 - \operatorname{erf}\sqrt{\psi}\right) + \frac{2}{\sqrt{\pi}}\sqrt{\psi}. \tag{17.68}$$

The error function has the small-argument limit

$$\lim_{z \to 0} \operatorname{erf} z \simeq \frac{2z}{\sqrt{\pi}} \tag{17.69}$$

and so for $\psi \ll 1$

$$\frac{n_i}{n_{i0}} = 1 + \psi, \tag{17.70}$$

which is identical to the Boltzmann result given by fluid theory.

However, because

$$\lim_{z \to \infty} \mathrm{e}^{-z^2} (1 - \operatorname{erf} z) = 0, \tag{17.71}$$

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Table 17.2

	Region 1	Region 2	Region 3
Definition	$\psi_d > \psi > 1$	$1 > \psi > 1/\bar{Z}$	$1/\bar{Z} > \psi$
Location	adjacent to dust grain	middle spherical layer	outer layer
Ions	non-Boltzmann, fast	Boltzmann	Boltzmann
Electrons	Boltzmann	Boltzmann	Boltzmann
Dust	zero density	zero density	Boltzmann
Dependence	vacuum-like w/const.	growing/decaying Yukawa	decaying Yukawa

when $\psi \gg 1$ the ion density has a non-Boltzmann dependence

$$\frac{n_i}{n_{i0}} = \frac{2}{\sqrt{\pi}}\sqrt{\psi}, \tag{17.72}$$

which is much smaller than the $\exp(\psi)$ dependence predicted by the Boltzmann relation.

Poisson's equation, Eq. (17.56), can be written non-dimensionally as

$$\begin{aligned}\frac{1}{\bar{r}^2}\frac{\partial}{\partial \bar{r}}\left(\bar{r}^2\frac{\partial \psi}{\partial \bar{r}}\right) &= \frac{n_i}{n_{i0}} - \frac{n_e}{n_{i0}} - \frac{Z_d n_d}{n_{i0}} \\ &= \frac{n_i}{n_{i0}} - (1-\alpha)\frac{n_e}{n_{e0}} - \alpha\frac{n_d}{n_{d0}},\end{aligned} \tag{17.73}$$

where, following the normalization convention introduced earlier, $\bar{r} = r/\lambda_{di}$. Upon substituting for the normalized densities, Poisson's equation becomes

$$\begin{aligned}\underbrace{\frac{1}{\bar{r}^2}\frac{\partial}{\partial \bar{r}}\left(\bar{r}^2\frac{\partial \psi}{\partial \bar{r}}\right)}_{\text{vacuum}} &= \underbrace{e^{\psi}\left(1-\text{erf}\left(\sqrt{\psi}\right)\right) + \frac{2}{\sqrt{\pi}}\sqrt{\psi}}_{\text{ions}} \\ &\quad -\underbrace{(1-\alpha)\exp\left(-\frac{\psi T_i}{T_e}\right)}_{\text{electrons}} \\ &\quad -\underbrace{\alpha\exp\left(-\bar{Z}\psi\right)}_{\text{dust}}.\end{aligned} \tag{17.74}$$

Because ψ becomes large near the dust grain, this equation is highly nonlinear and so the linearization technique used on p. 10 for the conventional Debye shielding derivation cannot be invoked. Instead, an approximate solution to Poisson's equation is obtained by separating the collisionless region around the dust grain test particle into three concentric layers (regions) according to the magnitude of ψ as shown in Table 17.2.

Poisson's equation has the following approximations in the three regions,

$$\text{Region 1}: \frac{1}{\bar{r}^2}\frac{\partial}{\partial \bar{r}}\left(\bar{r}^2\frac{\partial\psi}{\partial\bar{r}}\right) = 0 \tag{17.75}$$

$$\text{Region 2}: \frac{1}{\bar{r}^2}\frac{\partial}{\partial \bar{r}}\left(\bar{r}^2\frac{\partial\psi}{\partial\bar{r}}\right) = \psi + \alpha \tag{17.76}$$

$$\text{Region 3}: \frac{1}{\bar{r}^2}\frac{\partial}{\partial \bar{r}}\left(\bar{r}^2\frac{\partial\psi}{\partial\bar{r}}\right) = (1+\alpha\bar{Z})\psi. \tag{17.77}$$

The reasons for these approximations are as follows:

In Region 1, $\psi \gg 1$ so that the ion density is given in principle by Eq. (17.72) which would give a right-hand side scaling as $\sqrt{\psi}$. This ion term would dominate the electron and dust terms, since the latter are always less than unity in this region. Thus, keeping just the ion term on the right-hand side, Poisson's equation in Region 1 reduces to

$$\underbrace{\frac{1}{\bar{r}^2}\frac{\partial}{\partial \bar{r}}\left(\bar{r}^2\frac{\partial\psi}{\partial\bar{r}}\right)}_{\text{vacuum}} \simeq \underbrace{\frac{2}{\sqrt{\pi\psi}}\psi}_{\text{ion}}; \tag{17.78}$$

however, the ion term above can be neglected compared to the vacuum term because $\sqrt{\psi} \gg 1$ since $\psi \gg 1$. Thus Poisson's equation can be approximated in Region 1 by just the vacuum term. This assumption is quite good near the dust grain surface where ψ is indeed very large compared to unity, but becomes marginal when ψ approaches unity at the outer limit of Region 1.

In Region 2, defined by $1 > \psi > 1/\bar{Z}$, the ions have a Boltzmann distribution so $n_i/n_{e0} = 1+\psi$. The normalized electron density is $n_e/n_{e0} = \exp(-\psi T_i/T_e)$ and, since $T_i/T_e \ll 1$, this simplifies to $n_e/n_{e0} \simeq 1$. Also, because $\bar{Z}\psi \gg 1$ the dust density is nearly zero in this region.

In Region 3, defined by $1/\bar{Z} > \psi$, Eq. (17.74) approximates to

$$\begin{aligned}\underbrace{\frac{1}{\bar{r}^2}\frac{\partial}{\partial \bar{r}}\left(\bar{r}^2\frac{\partial\psi}{\partial\bar{r}}\right)}_{\text{vacuum}} &\simeq \underbrace{1+\psi}_{\text{ions}} - \underbrace{(1-\alpha)\left(1-\frac{\psi T_i}{T_e}\right)}_{\text{electrons}} - \underbrace{\left(1-\bar{Z}\psi\right)\alpha}_{\text{dust}} \\ &\simeq (1+\alpha\bar{Z})\psi,\end{aligned} \tag{17.79}$$

where $T_i/T_e \ll 1$ has been used. For finite α, the dust term dominates because $\alpha\bar{Z} \gg 1$.

The system has two boundary conditions, one at the dust grain surface and the other at infinity. The former occurs in Region 1 and is set by the radial electric field at the dust grain surface. This boundary condition is obtained from Gauss' law, which relates $\partial\psi/\partial\bar{r}$ at the dust grain surface to the dust grain charge and radius. The boundary condition at infinity can be considered as the large $\bar{r}$ limit for Region 3 and requires ψ to vanish as $\bar{r}$ goes to infinity.

If the same form of Poisson's equation were to characterize Regions 1, 2, and 3, then the system would be governed by a single second-order ordinary differential equation and the two boundary conditions described in the previous paragraph would suffice to give a unique solution. However, the equations in Regions 1, 2, and 3 differ from each other and so must be solved separately with appropriate matching at the two interfaces between the three regions. The criteria for matching at the interfaces, determined by integrating Poisson's equation twice across each interface, are that both ψ and its radial derivative must be continuous at each interface. An important feature of this analysis is that the locations of the two interfaces are unknowns, which are to be determined by solving the matching problem. Since Regions 1 and 2 are of finite extent, both decaying and non-decaying ψ solutions are allowed in these regions. However, only a decaying solution is allowed in Region 3.

Region 1, which is governed by Eq. (17.75), has vacuum-like solutions of the form $\psi \sim c\bar{r}^{-1} + d$, where c and d are constants. The constant c is chosen to give the correct radial electric field at the surface of the dust grain test charge and d is chosen to set $\psi = 1$ at $\bar{r}_i$, which is the as yet undetermined location of the interface between Regions 1 and 2. Thus, using Gauss' law to prescribe the radial electric field at the dust grain surface and choosing d to set $\psi = 1$ at $\bar{r} = \bar{r}_i$ gives

$$\psi_1(\bar{r}) = \frac{\dfrac{\alpha\bar{a}^3}{3} + \left(1 - \dfrac{1}{\bar{r}_i}\dfrac{\alpha\bar{a}^3}{3}\right)\bar{r}}{\bar{r}}. \tag{17.80}$$

In Region 2, which is governed by Eq. (17.76), the effective dependent variable is $\psi + \alpha$, which has growing and decaying Yukawa-like solutions $\sim \exp(\pm\bar{r})/\bar{r}$. It is convenient to write the Region 2 solution as

$$\psi_2(\bar{r}) = \frac{\bar{r}_o\left(\dfrac{1}{\bar{Z}} + \alpha\right)\cosh(\bar{r} - \bar{r}_o) + B\sinh(\bar{r} - \bar{r}_o) - \alpha\bar{r}}{\bar{r}}, \tag{17.81}$$

where the coefficients have been chosen so that $\psi = 1/\bar{Z}$ at $\bar{r}_o$, the as yet undetermined location of the interface between Regions 2 and 3. The coefficient B in this expression is undetermined for now, and it should be noted that ψ is independent of B at $\bar{r} = \bar{r}_o$.

Region 3, governed by Eq. (17.77), has a solution that can be expressed as

$$\psi_3(\bar{r}) = \frac{\bar{r}_0}{\bar{Z}\bar{r}}\exp\left(-\sqrt{\alpha\bar{Z} + 1}\,(\bar{r} - \bar{r}_o)\right) \tag{17.82}$$

where the coefficients have been chosen to give $\psi = 1/\bar{Z}$ at $\bar{r}_o$, the interface between Regions 2 and 3.

The condition that $\partial\psi/\partial\bar{r}$ is continuous at $\bar{r}_o$ gives

$$B = \alpha - \frac{\bar{r}_o}{\bar{Z}}\sqrt{\alpha\bar{Z}+1} \tag{17.83}$$

and this result completes the matching process at $\bar{r}_o$ since both ψ and $\partial\psi/\partial\bar{r}$ have now been arranged to be continuous at $\bar{r}_o$. Since $\psi_1 = 1$ has already been arranged at $\bar{r}_i$, all that is required to have continuity of ψ at $\bar{r}_i$ is to set $\psi_2 = 1$ at $\bar{r}_i$, i.e., set

$$\bar{r}_i = \bar{r}_o\left(\frac{1}{\bar{Z}}+\alpha\right)\cosh\left(\bar{r}_i-\bar{r}_o\right) + B\sinh\left(\bar{r}_i-\bar{r}_o\right) - \alpha\bar{r}_i. \tag{17.84}$$

What remains to be done is arrange for continuity of $\partial\psi/\partial\bar{r}$ at $\bar{r}=\bar{r}_i$. Since ψ has already been arranged to be continuous at $\bar{r}=\bar{r}_i$, continuity of $\partial\left(\bar{r}\psi\right)/\partial\bar{r}$ at $\bar{r}=\bar{r}_i$ implies continuity of $\partial\psi/\partial\bar{r}$. This means that continuity of $\partial\psi/\partial\bar{r}$ at $\bar{r}=\bar{r}_i$ is attained by equating the derivatives of the numerators of the right-hand sides of Eqs. (17.80) and (17.81) at $\bar{r}_i$, i.e., continuity of $\partial\psi/\partial\bar{r}$ at $\bar{r}=\bar{r}_i$ is attained by setting

$$1-\frac{1}{\bar{r}_i}\frac{\alpha\bar{a}^3}{3} = \bar{r}_o\left(\frac{1}{\bar{Z}}+\alpha\right)\sinh\left(\bar{r}_i-\bar{r}_o\right) + B\cosh\left(\bar{r}_i-\bar{r}_o\right) - \alpha. \tag{17.85}$$

Since $\bar{Z} \gg 1$ and α is of order unity, Eqs. (17.84) and (17.85) can be approximated as

$$(1+\alpha)\,\bar{r}_i = \alpha\bar{r}_o\cosh\left(\bar{r}_i-\bar{r}_o\right) + \alpha\sinh\left(\bar{r}_i-\bar{r}_o\right) \tag{17.86}$$

$$1+\alpha-\frac{1}{\bar{r}_i}\frac{\alpha\bar{a}^3}{3} = \alpha\bar{r}_o\sinh\left(\bar{r}_i-\bar{r}_o\right) + \alpha\cosh\left(\bar{r}_i-\bar{r}_o\right). \tag{17.87}$$

These constitute two coupled nonlinear equations in the unknowns $\bar{r}_i$ and $\bar{r}_o$ and can be solved numerically. Since the parameters α and $\bar{a}$ in these equations are functions of position in dusty plasma parameter space, Eqs. (17.86) and (17.87) can be solved for $\bar{r}_i$ and $\bar{r}_o$ at any point $\bar{a}, \bar{r}_d$ in dusty plasma parameter space. Because a different solution set $\{\bar{r}_i, \bar{r}_o\}$ exists at each point in dusty plasma parameter space, $\bar{r}_i$ and $\bar{r}_o$ may be considered as functions of position in dusty plasma parameter space, i.e., $\bar{r}_i = \bar{r}_i(\bar{a}, \bar{r}_d)$ and $\bar{r}_o = \bar{r}_o(\bar{a}, \bar{r}_d)$.

Now consider the density of the field dust grains in the vicinity of $\bar{r}=\bar{r}_o$. For $\bar{r} > \bar{r}_o$ the normalized potential is prescribed by Eq. (17.82) and decays at the dust Debye length, which is an extremely short length because $\bar{Z} \gg 1$. This precipitous spatial attenuation of ψ for $\bar{r} > \bar{r}_o$ means that dust grains are completely shielded from each other when their separation distance slightly exceeds $\bar{r}_o$. If this is the case, then the dust grains will not interact with each other; i.e., they can be considered as a gas of non-interacting particles. On the other hand, if $\bar{r} < \bar{r}_o$, the normalized potential ψ then exceeds $1/\bar{Z}$ and the dust grains will experience a strong mutual repulsion. This abrupt change-over is implicit in Eq. (17.61) which indicates that n_d/n_{d0} is near unity when $\psi \ll 1/\bar{Z}$ (i.e., just outside $\bar{r} = \bar{r}_o$) but n_d/n_{d0} is near zero when ψ becomes significantly larger than $1/\bar{Z}$ (i.e., just inside $\bar{r} = \bar{r}_o$). Hence, a dust grain test particle can be considered as a hard sphere of radius $\bar{r}_o$ when interacting with field dust grains.

Since the nominal separation between dust grains is $\bar{a}$, if $\bar{a} > \bar{r}_o$ then each dust grain is completely shielded from its neighbors, in which case the dust grains behave as an ideal gas of non-interacting particles. However, if $\bar{a} < \bar{r}_o$ is attempted, each dust grain experiences the full, unshielded repulsive force of its neighbors. This extreme repulsive force means that it is not possible for $\bar{a}$ to become less than $\bar{r}_o$. In effect, each dust grain sees its neighbor as a hard sphere of radius $\bar{r}_o$. Thus, when $\bar{a} = \bar{r}_o$ the collection of pressed-together dust grains should form a regularly spaced lattice structure with lattice spacing of order $\bar{a} = \bar{r}_o$. The dusty plasma has thus crystallized.

The condition for crystallization of a dusty plasma is thus

$$\bar{a} \leq \bar{r}_o(\bar{a}, \bar{r}_d), \tag{17.88}$$

which defines a curve in dusty plasma parameter space. Figure 17.4 shows a plot of contours of $\bar{r}_o/\bar{a}$ in dusty plasma parameter space; for reference, the region

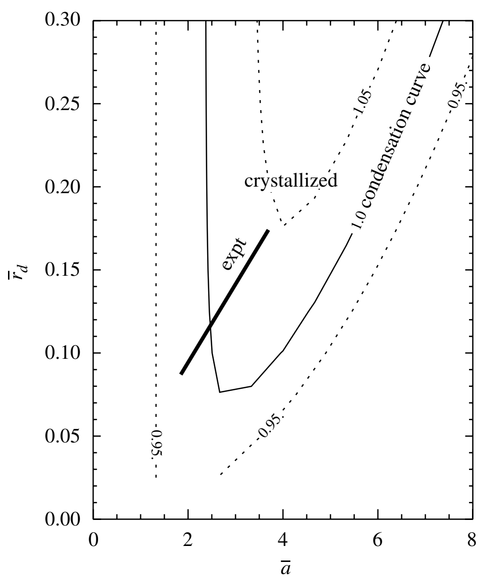

Fig. 17.4 Contours of $\bar{r}_o/\bar{a}$ as determined from solution to matching problem. Condensation (crystallization) is predicted to occur above the $\bar{r}_o/\bar{a} = 1$ line. Chu and I (1994) crystallization experiment, shown as bold line, is seen to lie in the condensation region.

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in parameter space associated with the Chu and I dusty plasma crystallization experiment is indicated as a bold line (as mentioned before, the finite length of this line corresponds to the density measurement error bars). The contour $\bar{r}_o(\bar{a}, \bar{r}_d)/\bar{a} = 1$ gives the crystallization condition; above this contour the dust grains are crystallized. It is also necessary to have $\bar{r}_d$ small as was specified in Eq. (17.13) so that ψ_d is not attenuated from its vacuum value by the effect of the shielding cloud.

Figure 17.5 shows plots of $\log_{10}\psi$, ψ, $10^4\psi$, n_e/n_{e0}, n_i/n_{i0}, and n_d/n_{d0} as functions of $\bar{r}$ for a dusty plasma on the verge of condensation. The functional

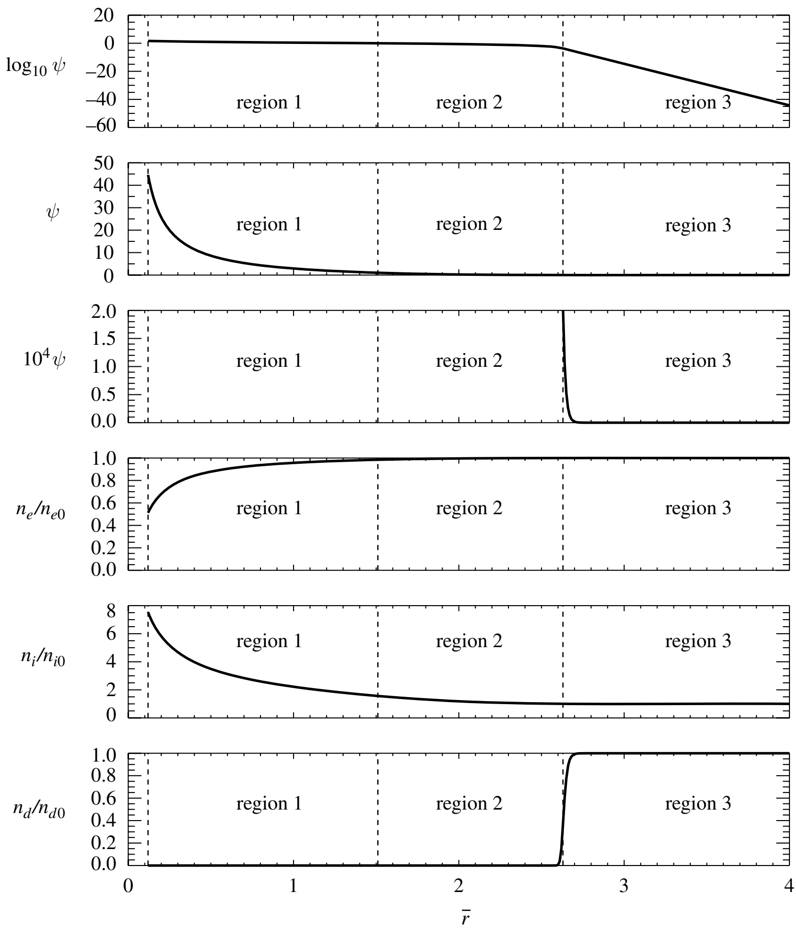

Fig. 17.5 Plots of $\log_{10}\psi$, ψ, $10^4\psi$, n_e/n_{e0}, n_i/n_{i0}, and n_d/n_{d0} as functions of $\bar{r}$ for a dusty plasma on the verge of condensation. Dust grain radius is at left of Region 1, parameters are from Bellan (2004b).

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form of ψ is obtained from the $\bar{r}_i$ and $\bar{r}_o$ values determined by solving Eqs. (17.86) and (17.87). The extremely rapid cutoff of the potential at $\bar{r}_o$, the interface between Regions 2 and 3, is evident and the characteristic scale of this cutoff is the dust Debye length. The normalized densities are plotted using Eqs. (17.60), (17.68), and (17.61). The field dust grain density vanishes for $\bar{r}$ inside $\bar{r}_o$, consistent with the hard-sphere behavior of the dust grain test particle. Although ψ is tiny at the interface between Regions 2 and 3, it nevertheless affects the average dust grain density profile at this location because of the extremely large charge on the dust grains.

17.8 Assignments

1. Assuming $T_e = 3\,\text{eV}$, $T_i = 0.03\,\text{eV}$, and $n_i = 10^9\ \text{cm}^{-3}$, estimate to within an order of magnitude how long it would take for $r = 5\,\mu\text{m}$ dust grains with density $n_d = 10^5$ cm^{-3} to become fully charged. Hint: estimate the initial electron current impinging on a dust grain and compare this to the final charge on the dust grain. Use Fig. 17.3 to estimate Z_d.
2. Orbital motion limit and angular momentum. The OML approximation ignores the effective potential resulting from centrifugal force. The validity of this approximation is examined here for the cases of an algebraically decaying central force and an exponentially decaying central force.
   (a) Show that when centrifugal force is taken into account, the radial equation of motion for a particle in a spherically symmetric electrostatic potential $\phi(r)$ is
   $$m\ddot{r} = -\frac{\partial \chi}{\partial r},$$
   where the effective potential χ is given by
   $$\chi = q\phi(r) + \frac{mb^2v^2}{2r^2}$$
   and m, b, and v are defined as in Section 17.2.
   (b) Show that if $q\phi$ is negative, a local minimum of the effective potential exists at some radial position r_1; compare this situation to planetary motion in the solar system.
   (c) Show that if $q\phi \sim -r^{-p}$ there can also be a local maximum at a radius r_2, where $r_2 > r_1$, providing p satisfies a certain condition. What is this condition for p? Plot $\chi(r)$ for the situation where there is a local maximum. Can a particle incident from infinity reach r_1 if its energy is less than $\chi(r_2)$? What sort of condition does this place on m, v, and b? Under what circumstances is the OML approximation valid? Plot trajectories for the situations where the OML approximation is valid and situations where it fails. Comment on whether OML is a reasonable approximation for dust charging.
   (d) Repeat (c) above for the situation where $q\phi \sim -\exp(-r/\lambda)$.

3. Assume the OML approximation is valid so that the effective potential can be considered to have a minimum in the vicinity of a negatively charged dust particle. Show that, if an incident ion does not hit the dust grain, it will reflect back to infinity. Show that a collision could cause an incident ion to be trapped in the effective potential well at r_1 as discussed in Assignment 1(b). Once an ion becomes trapped how long would it stay trapped (see Goree 1992)?
4. Dust Alfvén waves. Show that for waves with frequency below the dust grain cyclotron frequency, a dusty plasma will support MHD Alfvén waves with dispersion $\omega^2 = k_z^2 v_A^2$, where $v_A^2 = B^2/n_d m_d$.
5. Dust whistler waves. Show that a dusty plasma with $\alpha \simeq 1$ will support whistler-like waves with dispersion $\omega \simeq \omega_{ci} \cos\theta$, where $\omega_{cd} \ll \omega \ll \omega_{ci}$.
6. Make a plot like Fig. 17.3 for a dusty plasma with $T_e = T_i$ and discuss whether such a plasma could condense and form crystals.
7. Consider a methane plasma consisting of CH_4 ions and electrons. Assume that there exist some infinitesimal dust nuclei at $t = 0$ and that these become negatively charged as discussed in Section 17.2. Further assume that any methane ion hitting the dust grain sticks to the dust grain so that the mass of the dust grain increases with time. Plot the dust grain radius as a function of time. How long would it take for the dust grains to become sufficiently large to form a crystal (assume that $n_d = 10^5\ \text{cm}^{-3}$, $T_e/T_i = 100$, $\lambda_{di} = 100\,\mu\text{m}$)?
8. Consider a weakly ionized dusty plasma with $5\,\mu$m diameter dust grains, 3 eV electrons and room temperature (0.025 eV) argon ions. The dusty plasma is located above a horizontal metal plate lying in the $z = 0$ plane. Because electrons move faster than ions and dust grains, the electron flux to the metal plate is initially much larger than the ion or dust grain flux. The faster rate of loss for electrons causes the dusty plasma to become positively charged so that an electric field develops, which tends to retard the electrons and accelerate the ions towards the metal plate. Thus, the plasma potential is positive with respect to the metal plate and if the plasma potential is defined to be zero, then the metal plate has a negative potential, say ϕ_{plate}. The electron flux to the plate will thus be

$$\Gamma_e = n v_{Te} \exp\left(-q_e \phi_{plate}/\kappa T_e\right)$$

and, using a sheath analysis as in Section 2.9, the ion flux to the plate is

$$\Gamma_i = n c_s,$$

where $c_s = \sqrt{\kappa T_e/m_i}$ is the ion acoustic velocity. In equilibrium, the electron and ion fluxes will balance so that

$$\exp\left(-q_e \phi_{plate}/\kappa T_e\right) = \sqrt{\frac{m_e}{m_i}}$$

or

$$\phi_{plate} = -\frac{T_e}{2} \ln \frac{m_i}{m_e},$$

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where the temperature is expressed in electron volts. The change in potential will occur over a distance of the order of the electron Debye length going from the plate to the bulk plasma so that

$$\phi(z) = -\frac{T_e}{2} \exp\left(-z/\lambda_{De}\right) \ln \frac{m_i}{m_e},$$

where z is the distance above the metal plate. By considering the combined effect of (i) the vertical electric field associated with this potential and (ii) gravity acting on a dust grain, show that dust grains will tend to levitate above the metal plate. Plot the potential energy of dust grains in the combination of the electrostatic and gravitational potential fields. Assume that $Z_d = 10^4$ and that the dust grains are made of a material that has a mass density of 1 gm cc^{-1}.

9. Suppose a negatively charged dust grain has a charge Z_d and a radius r_d. Show that the potential on the surface of the dust grain is much less than $Z_d/4\pi\varepsilon_0 r_d$ if $r_d \gg \lambda_D$, where λ_D is the nominal shielding length for the shielding cloud around the dust grain. What implication does this have for dust charging theory and why do interesting dusty plasmas typically have $r_d \ll \lambda_D$?